Supplementary Online Content

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eTable 1. Patient Disposition by Site

Study Site Identifier	Screened Patients n (%) (N = 408)	Randomized Patients n (%) (N = 289)	Name and location of the Study Site
1001	180 (44.1)	137 (47.4)	Sun Yat-sen University Cancer Center, Guangzhou, China
1020	19 (4.7)	15 (5.2)	Affiliated Cancer Hospital & Institute of Guangzhou Medical University, Guangzhou, China
1021	14 (3.4)	11 (3.8)	Fudan University Cancer Center, Shanghai, China
1013	14 (3.4)	10 (3.5)	Affiliated Hospital of Guangdong Medical University, Zhanjiang, China
1015	16 (3.9)	9 (3.1)	Hunan Cancer Hospital and the Affiliated Cancer Hospital of Xiangya School of Medicine, Changsha, China
1005	14 (3.4)	9 (3.1)	Union Hospital Tongji Medical College Huazhong University of Science and Technology, Wuhan, China
1010	10 (2.5)	9 (3.1)	Jiangxi Cancer Hospital, Nanchang, China
1017	11 (2.7)	8 (2.8)	The Affiliated Cancer Hospital of Guizhou Medical University, Guiyang, China
1016	9 (2.2)	7 (2.4)	Shenzhen People's Hospital, Shenzhen, China
1019	9 (2.2)	6 (2.1)	The Second Xiangya Hospital of Central South University, Changsha, China
1014	9 (2.2)	6 (2.1)	The People’s Hospital of Guangxi Zhuang Autonomous Region, Nanning, China
2002	8 (2.0)	6 (2.1)	Taichung Veterans General Hospital, Taichung, Taiwan
1004	7 (1.7)	6 (2.1)	Fujian Provincial Cancer Hospital, Fuzhou, China
1018	6 (1.5)	6 (2.1)	West China Hospital of Sichuan University, Chengdu, China
1025	10 (2.5)	5 (1.7)	Jiangsu Cancer Hospital, Nanjing, China
2004	6 (1.5)	5 (1.7)	Chang Gung Memorial Hospital, Taoyuan, Taiwan
1007	5 (1.2)	5 (1.7)	Cancer Hospital of Shantou University Medical College, Shantou, China
3003	6 (1.5)	4 (1.4)	National Cancer Centre, Singapore City, Singapore
1012	8 (2.0)	3 (1.0)	Zhejiang Cancer Hospital, Hangzhou, China
1026	7 (1.7)	3 (1.0)	The Fifth Affiliated Hospital of Sun Yat-Sen University, Zhuhai, China
1003	5 (1.2)	3 (1.0)	Nanfang Hospital, Guangzhou, China
1011	4 (1.0)	3 (1.0)	Cancer Hospital Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing, China
1002	5 (1.2)	2 (0.7)	Tongji Hospital, Tongji Medical College, Huazhong, University of Science and Technology, Wuhan, China
1006	2 (0.5)	2 (0.7)	Shanghai General Hospital, Shanghai, China
1023	2 (0.5)	2 (0.7)	Fujian Medical University Union Hospital, Fuzhou, China
1024	3 (0.7)	1 (0.3)	Liuzhou Worker's Hospital, Liuzhou, China

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1022	3 (0.7)	1 (0.3)	Hainan General Hospital, Haikou, China
3002	3 (0.7)	1 (0.3)	Tan Tock Seng Hospital, Singapore City, Singapore
1008	3 (0.7)	1 (0.3)	The Fourth Hospital of Hebei Medical University Hebei Cancer Hospital, Shijiazhuang, China
2003	2 (0.5)	1 (0.3)	China Medical University Hospital, Taichung, Taiwan
2006	2 (0.5)	1 (0.3)	Taipei Veterans General Hospital, Taipei, Taiwan
2007	1 (0.2)	1 (0.3)	National Cheng Kung University Hospital, Tainan, Taiwan
1031	3 (0.7)	0	Guangxi Medical University Cancer Hospital
2001	1 (0.2)	0	National Taiwan University Hospital, Taiwan
1027	1 (0.2)	0	Guangzhou Panyu Central Hospital, Guangzhou, China

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eTable 2. Screening Failures: Common Reasons (≥5 Incidences) for not Meeting Inclusion Criteria or Meeting Exclusion Criteria

Code	Inclusion or exclusion criteria details	N
INCL9	Adequate organ function: · Hematologic: leucocytes ≥ 4.0 ×109/L, Absolute Neutrophil Count ≥ 2.0 × 109/L, hemoglobin ≥ 90 g/L, and platelets ≥ 100.0 ×109/L. · Hepatic: bilirubin ≤ 1.5 × upper limit of normal (ULN) (patients with known Gilbert's disease who have serum bilirubin level ≤ 3 × ULN may be enrolled), AST and ALT ≤ 3 ×ULN (AST/ALT ≤ 5 × ULN if liver metastases), with alkaline phosphatase ≤ 3 ×ULN (ALP ≤ 5 × ULN if liver or bone metastases); albumin ≥ 3 g/dL; · International Normalized Ratio (INR) or Prothrombin Time (PT) or Activated Partial Thromboplastin Time (aPTT) ≤ 1.5 × ULN. · Renal: serum creatinine ≤ 1.5 ULN and creatinine clearance ≥ 60 mL/min according to Cockcroft-Gault formula.	28
INCL4	Primarily metastatic (stage IVB as defined by the International Union against Cancer and American Joint Committee on Cancer staging system for NPC, eighth edition) or recurrent NPC after curative treatment, which is not amenable for local regional treatment or curative treatment. No previous systemic chemotherapy was given for the recurrent or metastatic disease.	16
EXCL19	Active infection including tuberculosis (clinical evaluation that includes clinical history, physical examination and radiographic findings, and TB testing in line with local practice),hepatitis B (known positive HBV surface antigen (HBsAg) result and HBV DNA detected by study site lab > 1000 cps/mL or the lower limit of the local lab), hepatitis C, or human immunodeficiency virus (positive HIV 1/2 antibodies). Patients with a past or resolved HBV infection (defined as the presence of hepatitis B core antibody [anti-HBc] and absence of HBsAg) are eligible only if they are negative for HBVDNA (HBV DNA detected by study site lab < 1000 cps/mL or the lower limit of the local lab). Patients positive for hepatitis C (HCV) antibody are eligible only if polymerase chain reaction is negative for HCV RNA.	15
INCL11	Willingness and ability to comply with scheduled visits, treatment plans, laboratory tests and other study procedures	12
INCL5	For the recurrent NPC after curative treatment (including radiotherapy and/or induction, concurrent or adjuvant chemotherapy), the interval between recurrence and the last dose of previous radiotherapy or chemotherapy must be more than 6 months.	10
INCL6	At least 1 measurable lesion according to RECIST version 1.1.	6

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eTable 3. Summary of Tumor Biopsies. Biopsies taken within 6 weeks of the first dose of the study treatment were considered fresh biopsies, whereas biopsies taken more than 6 weeks of the first dose of the study treatment were considered archival samples. Among 59 metastatic tumor biopsies, 29 were from metastatic lymph nodes.

	Number of patients N (%)	Fresh Samples N (%)	Archival Samples N (%)
Primary tumors	220 (76.1%)	106 (36.7%)	114 (39.4%)
Metastatic sites	59 (20.4%)	44 (15.2%)	15 (5.2%)
No biopsy collected	10 (3.5%)	-	-

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eTable 4. Extent of Study Drug Exposure as of May 8, 2022

	Toripalimab +chemotherapy (N=146)	**Placebo +chemotherapy (N=143)**
The number of toripalimab/Placebo cycles		
Mean	19.9	12.7
Sd	12.15	6.75
Median	20.0	12.0
Minimum, Maximum	1, 35	1, 31
Weeks of exposure for toripalimab/Placebo		
Mean	64.40	41.54
Sd	38.414	21.748
Median	65.70	37.30
Minimum, Maximum	3.1, 107.4	3.0, 101.1
The number of cycles cisplatin received		
Mean	5.1	5.3
Sd	1.50	1.44
Median	6.0	6.0
Minimum, Maximum	1, 6	1, 6
Weeks of exposure for cisplatin		
Mean	17.20	17.22
Sd	5.662	4.932
Median	18.30	18.40
Minimum, Maximum	3.0, 33.1	3.0, 28.1
The number of cycles gemcitabine received		
Mean	5.3	5.4
Sd	1.34	1.35
Median	6.0	6.0
Minimum, Maximum	1, 6	1, 6
Weeks of exposure for gemcitabine		
Mean	18.66	18.59
Sd	5.304	4.723
Median	19.30	19.60
Minimum, Maximum	3.0, 34.1	3.0, 29.1

Sd, standard deviation.

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eTable 5. Subsequent Anti-Cancer Therapy in Intent-to-Treat Population After the Study Treatment

	Toripalimab + Chemo N = 146 (%)	**Placebo + Chemo N= 143 (%)**	**Total N=289 (%)**
New anti-cancer therapy	82 (56.2)	106 (74.1)	188 (65.1)
Systemic treatment	77 (52.7)	105 (73.4)	182 (63)
Radiotherapy	18 (12.3)	23 (16.1)	41 (14.2)
Systemic treatment	77 (52.7)	105 (73.4)	182 (63)
Cytotoxic Therapy	55 (37.7)	99 (69.2)	154 (53.3)
PD-1/L1 Inhibitors	48 (32.9)	49 (34.3)	97 (33.6)
CAMRELIZUMAB	21 (14.4)	24 (16.8)	45 (15.6)
TORIPALIMAB	14 (9.6)	17 (11.9)	31 (10.7)
SINTILIMAB	8 (5.5)	9 (6.3)	17 (5.9)
TISLELIZUMAB	7 (4.8)	4 (2.8)	11 (3.8)
QL1706	3 (2.1)	3 (2.1)	6 (2.1)
SHR1701	3 (2.1)	1 (0.7)	4 (1.4)
PEMBROLIZUMAB	0	3 (2.1)	3 (1.0)
PENPULIMAB	0	2 (1.4)	2 (0.7)
NIVOLUMAB	1 (0.7)	1 (0.7)	2 (0.7)
DURVALUMAB	0	1 (0.7)	1 (0.3)
AVELUMAB	0	1 (0.7)	1 (0.3)
Protein Kinase Inhibitor	28 (19.2)	38 (26.6)	66 (22.8)
EGFR antibody	7 (4.8)	19 (13.3)	26 (9.0)
NIMOTUZUMAB	4 (2.7)	17 (11.9)	21 (7.3)
CETUXIMAB	2 (1.4)	1 (0.7)	3 (1.0)
MRG003	1 (0.7)	1 (0.7)	2 (0.7)
Other immunomodulators	2 (1.4)	8 (5.6)	10 (3.5)
Anti-angiogenic Agent (non-TKI)	3 (2.1)	2 (1.4)	5 (1.7)
Others	9 (6.2)	4 (2.8)	13 (4.5)
Unknown	1 (0.7)	1 (0.7)	2 (0.7)

Data Cutoff: Nov 18, 2022

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eTable 6. Overall Survival in Intent-to-Treat Population as of November 18, 2022

	Toripalimab +chemotherapy (N=146)	**Placebo + chemotherapy (N=143)**	**Total (N=289)**
Status, n (%)			
Death	57 (39.0)	76 (53.1)	133 (46.0)
Censored	89 (61.0)	67 (46.9)	156 (54.0)
Alive	82 (56.2)	60 (42.0)	142 (49.1)
Lost to Follow-up	7 (4.8)	7 (4.9)	14 (4.8)
Unknown	0	0	0
Estimate of median OS (months)	NE	33.7	41.8
(95% CI)[1]	(38.70, NE)	(27.01, 44.19)	(36.57, NE)
Stratified analyses[2]			
Estimate of HR (toripalimab vs Placebo)[3] (95% CI)	0.63 (0.446, 0.891)		
Log-rank p-value (two-sided)	0.008		
Unstratified analyses			
Estimate of HR (toripalimab vs Placebo)[3] (95% CI)	0.66 (0.465, 0.925)		
Log-rank p-value (two-sided)	0.02		
1-Year OS Rate	90.9%	87.1%	89.1%
(95% CI)[4]	(84.87, 94.62)	(80.36, 91.69)	(84.80, 92.17)
Difference of 1-Year OS Rates	3.8%		
(95% CI)[5]	(-3.5, 11.1)		
2-Year OS Rate	78.0%	65.1%	71.6%
(95% CI)[4]	(70.18, 83.97)	(56.50, 72.44)	(65.93, 76.55)
Difference of 2-Year OS Rates	12.9%		
(95% CI)[5]	(2.3, 23.4)		
3-Year OS Rate	64.5%	49.2%	56.9%
(95% CI)[4]	(55.86, 71.87)	(40.53, 57.32)	(50.83, 62.58)
Difference of 3-Year OS Rates	15.3%		
(95% CI)[5]	(3.6, 26.9)		

NE, not estimable.

Note: OS is defined as the time from randomization to death from any cause and is calculated as (death date or censoring date - randomization date + 1)/30.4375 in month.

[1] The Brookmeyer Crowley methodology was used to construct the 95% CI for the median overall survival. [2] The stratified analyses used ECOG performance status (0 vs. 1), and disease stage (recurrent vs. metastatic) as recorded in the IWRS.

[3] The hazard ratio was estimated with the use of the Cox proportional hazards model. Efron's method was used to handle ties.

[4] The 95% CI was estimated with use of the standard error derived from Greenwood's formula.

[5] The 95% CI was estimated with use of the normal approximation method.

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eTable 7. PFS and OS Treatment Effects in Different PD-L1 Expression Subgroups. Archival or fresh tumor biopsy samples were obtained from patients prior to treatment. PD-L1 expression was evaluated by immunohistochemistry (IHC) staining using a validated JS311 assay in a central lab. Three different PD-L1 expression cutoffs were evaluated, including PD-L1 tumor cell (TC) or immune cell (IC) ≥1% vs. TC ≥1% vs. IC ≥1%. Three cutoffs had similar OS and PFS treatment effects in respective subgroups. NE, not estimable.

		PFS		OS	
Cutoffs for PD-L1 Status	**No. of Patients**	**Hazard Ratio (95% CI)**	**Median in month, Toripalimab vs Placebo**	**Hazard Ratio (95% CI)**	**Median in month, Toripalimab vs Placebo**
TC ≥ 1% or IC ≥ 1%	218	0.59 (0.405, 0.859)	19.4 vs. 8.3	0.71 (0.480, 1.043)	NE vs. 35.2
TC < 1% and IC < 1%	45	0.37 (0.168, 0.799)	NE vs. 6.8	0.36 (0.138, 0.953)	NE vs. 24.6
TC ≥ 1%	190	0.66 (0.444, 0.989)	15.4 vs.9.5	0.76 (0.503, 1.148)	NE vs. 33.7
TC < 1%	73	0.32 (0.165, 0.622)	NE vs. 7.0	0.37 (0.171, 0.800)	NE vs. 30.3
IC ≥ 1%	196	0.65 (0.433, 0.963)	19.4 vs. 9.5	0.76 (0.503, 1.140)	NE vs. 37.6
IC < 1%	67	0.32 (0.163, 0.610)	21.2 vs. 6.8	0.35 (0.162, 0.775)	NE vs. 24.6

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eTable 8. Overview of Treatment-Emergent Adverse Events (Safety Population)

Category	Toripalimab +chemotherapy (N=146) n (%)	Placebo +chemotherapy (N=143) n (%)	Total (N=289) n (%)
Any TEAE	146 (100.0)	143 (100.0)	289 (100.0)
Grade ≥3	131 (89.7)	129 (90.2)	260 (90.0)
Any irAE	79 (54.1)	31 (21.7)	110 (38.1)
Any irAE with Grade ≥3	14 (9.6)	2 (1.4)	16 (5.5)
Any AESI [1]	1 (0.7)	1 (0.7)	2 (0.7)
Any AESI with Grade ≥3	1 (0.7)	1 (0.7)	2 (0.7)
Any IRR	6 (4.1)	6 (4.2)	12 (4.2)
Any IRR with Grade ≥3	0	0	0
Any SAE	64 (43.8)	62 (43.4)	126 (43.6)
Any SAE which is life-threatening	5 (3.4)	6 (4.2)	11 (3.8)
Any SAE requires in-patient hospitalization or prolongs existing hospitalization	60 (41.1)	57 (39.9)	117 (40.5)
Any SAE results in persistent or significant disability/incapacity	0	0	0
Any SAE which is congenital anomaly/birth defect	0	0	0
Any SAE which is other medically important condition	6 (4.1)	3 (2.1)	9 (3.1)
Any TEAE related to study drug	141 (96.6)	139 (97.2)	280 (96.9)
Grade ≥3	119 (81.5)	120 (83.9)	239 (82.7)
SAE	56 (38.4)	52 (36.4)	108 (37.4)
Any TEAE related to gemcitabine	146 (100.0)	143 (100.0)	289 (100.0)
Grade ≥3	127 (87.0)	129 (90.2)	256 (88.6)
SAE	53 (36.3)	57 (39.9)	110 (38.1)
Any TEAE related to cisplatin	145 (99.3)	143 (100.0)	288 (99.7)
Grade ≥3	126 (86.3)	128 (89.5)	254 (87.9)
SAE	56 (38.4)	57 (39.9)	113 (39.1)
Any TEAE leading to drug withdrawal of study drug	17 (11.6)	7 (4.9)	24 (8.3)
SAE	14 (9.6)	5 (3.5)	19 (6.6)

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Any TEAE leading to drug withdrawal of gemcitabine	20 (13.7)	16 (11.2)	36 (12.5)
SAE	12 (8.2)	11 (7.7)	23 (8.0)
Any TEAE leading to drug withdrawal of cisplatin	27 (18.5)	19 (13.3)	46 (15.9)
SAE	13 (8.9)	9 (6.3)	22 (7.6)
Any TEAE leading to drug interruption of study drug	74 (50.7)	69 (48.3)	143 (49.5)
Any TEAE leading to drug interruption of gemcitabine	104 (71.2)	103 (72.0)	207 (71.6)
Any TEAE leading to drug interruption of cisplatin	63 (43.2)	49 (34.3)	112 (38.8)
Any TEAE leading to dose reduction of gemcitabine	71 (48.6)	74 (51.7)	145 (50.2)
Any TEAE leading to dose reduction of cisplatin	59 (40.4)	62 (43.4)	121 (41.9)
Any TEAE with outcome of death (fatal)	5 (3.4)	4 (2.8)	9 (3.1)
SAE	5 (3.4)	4 (2.8)	9 (3.1)

Data cut-off date: May 08, 2022

AESI, adverse events of special interest; irAE, immune-related adverse event; IRR, infusion-related reaction; TEAE, treatment-emergent adverse event.

Note: TEAEs are adverse events that either start or worsen in severity on or after the date/time of first dose of study treatment and on or before 60 days after the date/time of last dose of study treatment.

[1] Adverse events of special interest (AESI) included suspected immune-related myocarditis and abnormal liver function which meet the criteria of Hy's law, which had not been reported as serious adverse events, as collected in CRF.

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eTable 9. Summary of Treatment-Emergent Adverse Events (TEAE) Occurring in at Least 5% of Patients in Either Study Arm

TEAE N (%)	**Toripalimab (N=146)**	**Placebo (N=143)**
Any TEAE	146 (100.0)	143 (100.0)
Leukopenia	133 (91.1)	135 (94.4)
Anemia	130 (89.0)	135 (94.4)
Neutropenia	126 (86.3)	133 (93.0)
Nausea	103 (70.5)	121 (84.6)
Vomiting	99 (67.8)	94 (65.7)
Thrombocytopenia	94 (64.4)	88 (61.5)
Decreased appetite	81 (55.5)	90 (62.9)
Aspartate aminotransferase increased	58 (39.7)	45 (31.5)
Constipation	58 (39.7)	66 (46.2)
Alanine aminotransferase increased	56 (38.4)	57 (39.9)
Fatigue	54 (37.0)	54 (37.8)
Hypothyroidism	53 (36.3)	25 (17.5)
Rash	51 (34.9)	39 (27.3)
Pyrexia	47 (32.2)	35 (24.5)
Diarrhea	45 (30.8)	33 (23.1)
Neuropathy peripheral	45 (30.8)	45 (31.5)
Cough	41 (28.1)	38 (26.6)
Hyponatremia	38 (26.0)	53 (37.1)
Upper respiratory tract infection	37 (25.3)	20 (14.0)
Musculoskeletal pain	37 (25.3)	36 (25.2)
Hypokalemia	35 (24.0)	36 (25.2)
Insomnia	35 (24.0)	26 (18.2)
Stomatitis	34 (23.3)	27 (18.9)
Dizziness	32 (21.9)	31 (21.7)
Blood creatinine increased	28 (19.2)	34 (23.8)
Hypochloremia	27 (18.5)	37 (25.9)
Headache	27 (18.5)	32 (22.4)
Pneumonia	26 (17.8)	10 (7.0)
Pruritus	25 (17.1)	11 (7.7)
Abdominal pain	24 (16.4)	15 (10.5)
Hypoproteinemia	17 (11.6)	17 (11.9)
Lymphopenia	17 (11.6)	16 (11.2)
Abdominal distension	16 (11.0)	10 (7.0)
Weight decreased	16 (11.0)	13 (9.1)
Hypertension	15 (10.3)	8 (5.6)
Dry mouth	15 (10.3)	11 (7.7)
Hyperuricemia	15 (10.3)	16 (11.2)
Epistaxis	14 (9.6)	19 (13.3)

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Hepatic function abnormal	14 (9.6)	10 (7.0)
Rhinorrhea	13 (8.9)	10 (7.0)
Toothache	12 (8.2)	3 (2.1)
Hypomagnesaemia	11 (7.5)	20 (14.0)
Oedema	11 (7.5)	17 (11.9)
Chest discomfort	10 (6.8)	11 (7.7)
Thyroid function test abnormal	10 (6.8)	9 (6.3)
Tinnitus	9 (6.2)	12 (8.4)
Gamma-glutamyl transferase increased	8 (5.5)	12 (8.4)
Hematuria	8 (5.5)	4 (2.8)
Hyperglycemia	8 (5.5)	7 (4.9)
Thrombosis	7 (4.8)	12 (8.4)
Alopecia	7 (4.8)	11 (7.7)
Creatinine renal clearance decreased	7 (4.8)	9 (6.3)
Hypocalcemia	5 (3.4)	16 (11.2)
Otitis media	4 (2.7)	8 (5.6)

Data cut-off date: May 08, 2022.

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eTable 10. Summary of ≥Grade 3 Treatment-Emergent Adverse Events (TEAE) Occurring in at Least 1% of Patients in Either Study Arm

TEAE N (%)	**Toripalimab (N=146)**	**Placebo (N=143)**
Any TEAE with Grade >=3	131 (89.7)	129 (90.2)
Leukopenia	90 (61.6)	84 (58.7)
Neutropenia	86 (58.9)	91 (63.6)
Anemia	72 (49.3)	58 (40.6)
Thrombocytopenia	49 (33.6)	41 (28.7)
Pneumonia	17 (11.6)	5 (3.5)
Hyponatremia	13 (8.9)	6 (4.2)
Lymphopenia	13 (8.9)	10 (7.0)
Hypokalemia	12 (8.2)	11 (7.7)
Hypertension	9 (6.2)	6 (4.2)
Rash	5 (3.4)	3 (2.1)
Upper respiratory tract infection	5 (3.4)	4 (2.8)
Syncope	4 (2.7)	2 (1.4)
Electrolyte imbalance	3 (2.1)	3 (2.1)
Epistaxis	3 (2.1)	4 (2.8)
Fatigue	3 (2.1)	3 (2.1)
Hepatic function abnormal	3 (2.1)	4 (2.8)
Hypomagnesaemia	3 (2.1)	6 (4.2)
Vomiting	3 (2.1)	3 (2.1)
Alanine aminotransferase increased	2 (1.4)	0
Aspartate aminotransferase increased	2 (1.4)	2 (1.4)
Bone marrow failure	2 (1.4)	5 (3.5)
Diarrhea	2 (1.4)	0
Disease progression	2 (1.4)	3 (2.1)
Encephalopathy	2 (1.4)	0
Hypochloremia	2 (1.4)	0
Hypoglycemia	2 (1.4)	0
Nausea	2 (1.4)	4 (2.8)
Pyrexia	2 (1.4)	1 (0.7)

Data cut-off date: May 08, 2022.

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eTable 11. Summary of Immune-Related Treatment-Emergent Adverse Events Determined by Investigator Occurring in at Least 1% of Patients in Either Study Arm

irAE N (%)	**Toripalimab (N=146)**	**Placebo (N=143)**
Any irAE	79 (54.1)	31 (21.7)
Hypothyroidism	34 (23.3)	12 (8.4)
Rash	21 (14.4)	9 (6.3)
Pruritus	12 (8.2)	6 (4.2)
Alanine aminotransferase increased	10 (6.8)	2 (1.4)
Aspartate aminotransferase increased	10 (6.8)	2 (1.4)
Thyroid function test abnormal	8 (5.5)	4 (2.8)
Blood creatinine increased	5 (3.4)	1 (0.7)
Thrombocytopenia	5 (3.4)	2 (1.4)
Pyrexia	5 (3.4)	1 (0.7)
Anemia	4 (2.7)	1 (0.7)
Fatigue	4 (2.7)	0
Hyperthyroidism	4 (2.7)	0
Pneumonia	3 (2.1)	0
Leukopenia	3 (2.1)	3 (2.1)
Electrolyte imbalance	2 (1.4)	0
Neutropenia	2 (1.4)	2 (1.4)
Arthritis	2 (1.4)	0
Blood creatine phosphokinase increased	2 (1.4)	0
Hematuria	2 (1.4)	0
Hyperuricemia	2 (1.4)	1 (0.7)
Neuropathy peripheral	2 (1.4)	1 (0.7)
Breast pain	1 (0.7)	2 (1.4)

Data cut-off date: May 08, 2022.

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eTable 12. Summary of ≥Grade 3 Immune-Related Treatment-Emergent Adverse Events Determined by Investigator

irAE N (%)	**Toripalimab (N=146)**	**Placebo (N=143)**
Any irAE with Grade >=3	14 (9.6)	2 (1.4)
Rash	5 (3.4)	1 (0.7)
Thrombocytopenia	3 (2.1)	0
Pneumonia	2 (1.4)	0
Anemia	1 (0.7)	0
Blood creatinine increased	1 (0.7)	0
Electrolyte imbalance	1 (0.7)	0
Encephalopathy	1 (0.7)	0
Hepatic function abnormal	1 (0.7)	0
Hypoglycemia	1 (0.7)	0
Hypothyroidism	1 (0.7)	0
Immune-mediated renal disorder	1 (0.7)	0
Metabolic acidosis	1 (0.7)	0
Myocarditis	1 (0.7)	0
Myositis	1 (0.7)	0
Neutropenia	1 (0.7)	1 (0.7)
Psoriasis	1 (0.7)	0
Leukopenia	0	1 (0.7)

Data cut-off date: May 08, 2022.

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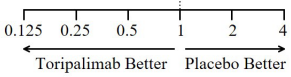
SUPPLEMENTAL FIGURES

Subgroup	Toripalimab n events / N total	Placebo n events / N total	Hazard Ratio (95% CI)	P Value	P Interaction
Progression-Free Survival (Primary Endpoint)					
Overall	63 / 146	87 / 143	0.52 (0.374, 0.726)	<0.001	
Age (years)					0.09
≤ 50	38 / 97	44 / 69	0.43 (0.278, 0.672)	<0.001	
> 50	25 / 49	43 / 74	0.71 (0.434, 1.174)	0.18	
Gender					0.96
Female	12 / 22	17 / 27	0.50 (0.240, 1.063)	0.06	
Male	51 / 124	70 / 116	0.53 (0.368, 0.765)	<0.001	
Baseline disease stage					0.27
Recurrent	31 / 85	51 / 87	0.45 (0.287, 0.714)	<0.001	
Metastatic	32 / 61	36 / 56	0.60 (0.370, 0.977)	0.04	
Disease-Free Interval (years)					0.27
≤ 2	18 / 48	29 / 45	0.36 (0.197, 0.650)	<0.001	
> 2	9 / 27	15 / 34	0.59 (0.247, 1.401)	0.23	
Baseline ECOG					0.42
0	40 / 83	52 / 81	0.57 (0.376, 0.869)	0.008	
1	23 / 63	35 / 62	0.46 (0.273, 0.793)	0.004	
Baseline PD-L1 expression level					0.17
TC ≥ 1% or IC ≥ 1%	49 / 109	64 / 109	0.59 (0.405, 0.859)	0.006	
TC < 1% and IC < 1%	10 / 21	19 / 24	0.37 (0.168, 0.799)	0.009	
Baseline EBV copy number					0.22
< 2000	20 / 54	24 / 54	0.70 (0.387, 1.274)	0.24	
≥ 2000	43 / 92	63 / 89	0.42 (0.283, 0.635)	<0.001	
Overall Survival (Secondary Endpoint)					
Overall	57 / 146	76 / 143	0.63 (0.446, 0.891)	0.008	
Age (years)					0.98
≤ 50	34 / 97	32 / 69	0.70 (0.433, 1.138)	0.15	
> 50	23 / 49	44 / 74	0.69 (0.418, 1.148)	0.15	
Gender					0.81
Female	9 / 22	15 / 27	0.62 (0.269, 1.410)	0.25	
Male	48 / 124	61 / 116	0.67 (0.460, 0.981)	0.04	
Baseline disease stage					0.004
Recurrent	29 / 85	55 / 87	0.44 (0.279, 0.688)	<0.001	
Metastatic	28 / 61	21 / 56	1.23 (0.701, 2.175)	0.46	
Disease-Free Interval (years)					0.38
≤ 2	17 / 48	33 / 45	0.38 (0.209, 0.678)	<0.001	
> 2	10 / 27	19 / 34	0.58 (0.271, 1.258)	0.17	
Baseline ECOG					0.21
0	28 / 83	34 / 81	0.80 (0.483, 1.313)	0.37	
1	29 / 63	42 / 62	0.51 (0.315, 0.817)	0.005	
Baseline PD-L1 expression level					0.18
TC ≥ 1% or IC ≥ 1%	45 / 109	59 / 109	0.71 (0.480, 1.043)	0.08	
TC < 1% and IC < 1%	6 / 21	14 / 24	0.36 (0.138, 0.953)	0.03	
Baseline EBV copy number					0.56
< 2000	16 / 54	21 / 54	0.76 (0.397, 1.462)	0.41	
≥ 2000	41 / 92	55 / 89	0.59 (0.396, 0.890)	0.01	



eFigure 1. Treatment Effects in Key Subgroups. The final PFS (Top Panel) and OS (Bottom Panel) treatment effects are explored in key subgroups. PD-L1 positivity was defined as the presence of membrane staining of any intensity in ≥1% of tumor cells (TC) or immune cells (IC) by JS311 staining. Plasma EBV DNA copy number was determined by qRT-PCR method with probes against EBV genes in a central lab.

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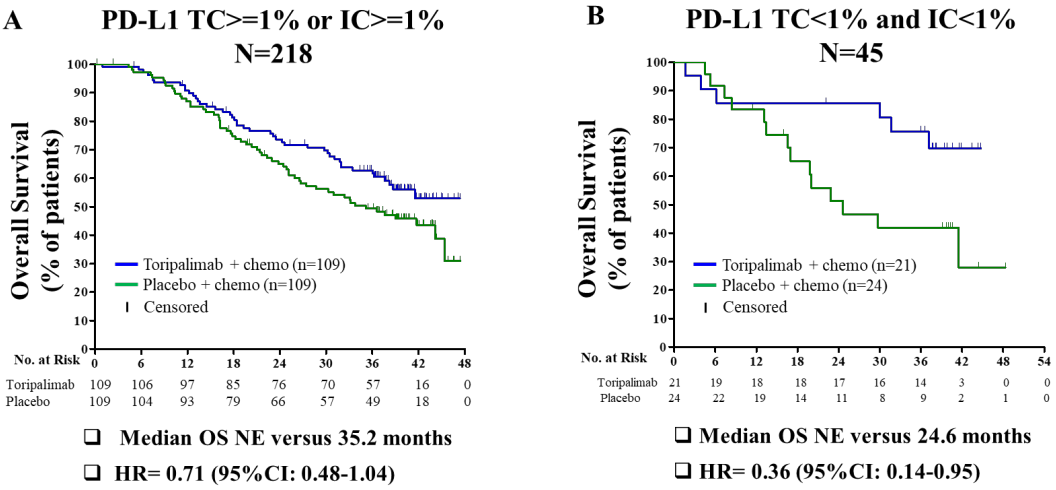

eFigure 2. Final Overall Survival Analysis in the PD-L1+ (A) and PD-L1- (B) Subgroups. PD-L1 positivity was defined as the presence of membrane staining of any intensity in ≥1% of tumor cells (TC) or immune cells (IC) by JS311 staining. Kaplan-Meier estimates of OS are shown to compare the toripalimab plus chemotherapy arm with the placebo plus chemotherapy arm. Censored patients are marked with " ❙ " in the graph. The number of patients at risk at indicated time points are shown below the x-axis. The median OS, and stratified hazard ratio for death are shown under the Kaplan-Meier curves.

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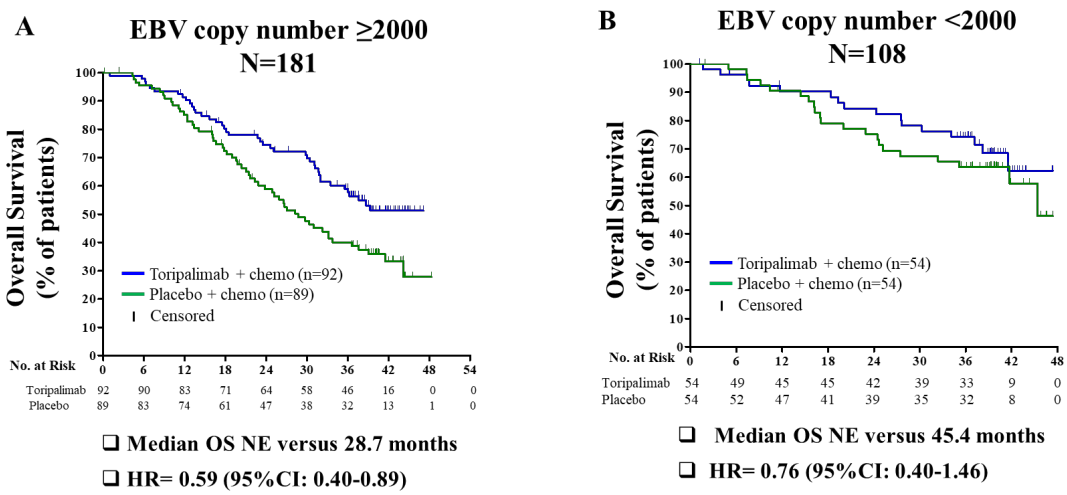

eFigure 3. Final Overall Survival Analysis in EBV Copy Number ≥2000 IU/mL Subgroup (A) and <2000 IU/mL Subgroup (B). Plasma EBV DNA copy number was determined in a centralized laboratory by real-time fluorescence quantitative PCR detection technology with a pair of specific primers designed for the conserved region of EBV nucleic acid. Kaplan-Meier estimates of OS are shown to compare the toripalimab plus chemotherapy arm with the placebo plus chemotherapy arm. Censored patients are marked with " ❙ " in the graph. The number of patients at risk at indicated time points are shown below the x-axis. The median OS, and stratified hazard ratio for death are shown under the Kaplan-Meier curves.

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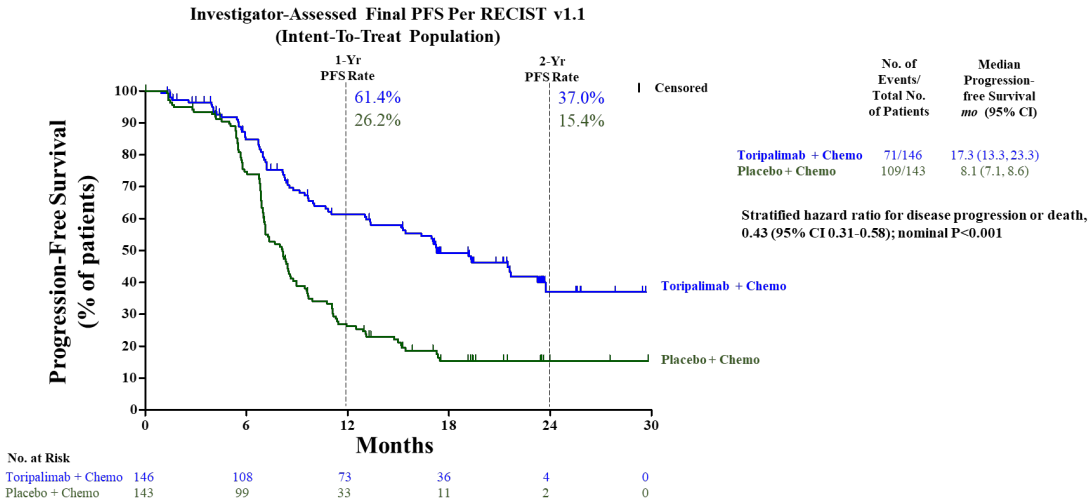

eFigure 4. Final Progression-free Survival Analysis in the Intention-to-Treat Population by Investigator. Kaplan-Meier-estimated PFS curves as assessed by investigator according to RECIST v1.1. Censored patients are marked with " ❙ " in the graph. The number of patients at risk at indicated time points shown below x-axis. One- and two-year PFS rates are shown on the graph. The number of events, median PFS, and stratified hazard ratio for PFS shown to the right of Kaplan-Meier curves. The cutoff date for the final PFS analysis is June 8, 2021.

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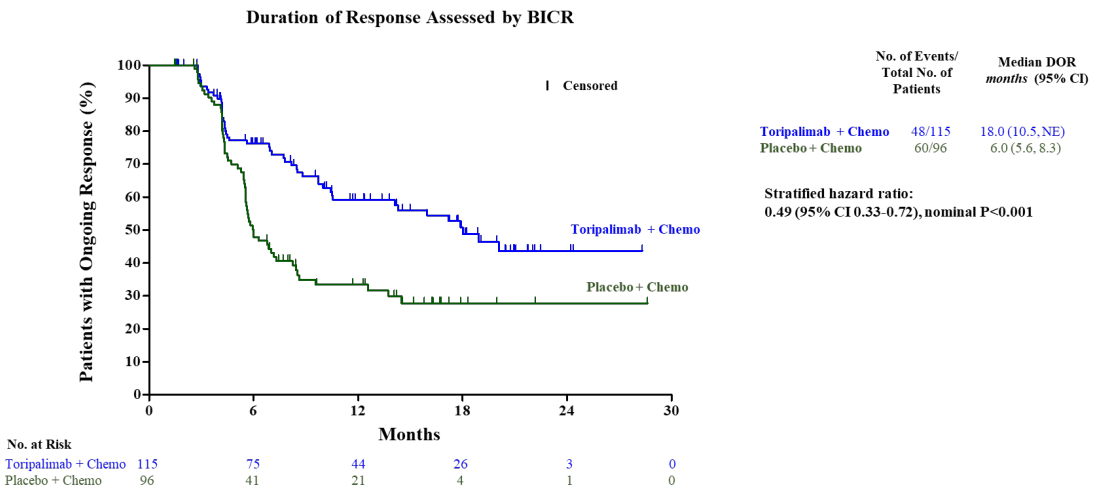

eFigure 5. Duration of Response by Blinded Independent Central Review per RECIST v1.1. Kaplan-Meier-estimated DoR curves as assessed by BICR according to RECIST v1.1. Censored patients are marked with " ❙ " in the graph. The number of patients at risk at indicated time points shown below x-axis. The number of events, median DoR, and stratified hazard ratio for DoR were shown to the right of Kaplan-Meier curves. The cutoff date for DoR analysis is June 8, 2021.

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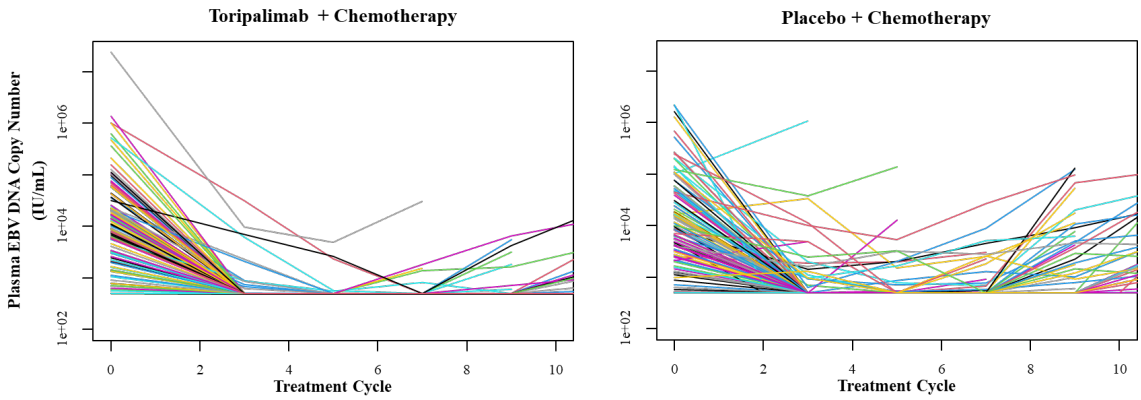

eFigure 6. Dynamic Plasma EBV DNA Copy Numbers During the Study.

Longitudinal plasma samples were collected for dynamic EBV DNA copy number evaluation, including samples from baseline and multiple treatments cycles specified in the study protocol. Plasma EBV DNA copy number was determined by qRT-PCR method with probes against EBV genes in a central lab. The detection limit was 500 IU/mL. The EBV DNA copy number in the toripalimab group (LEFT) and the placebo group (RIGHT) over treatment cycles are shown.

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eMethods

PATIENTS

Patients must have at least 1 measurable lesion according to the Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1; Eastern Cooperative Oncology Group (ECOG) performance status score of 0 or 1; and adequate organ function. Key exclusion criteria included a history of severe hypersensitivity reactions to the study treatment or any other monoclonal antibody; active or untreated CNS metastases; prior anti-PD-1/PD-L1/CTLA4 treatment; a history of bone marrow or solid organ transplantation; and a history of protocol specified autoimmune diseases. The full eligibility criteria are available in the study protocol.

RANDOMIZATION

Patient randomization and dispensing of investigational drugs were implemented with Interactive Web Response System (IWRS) using the stratification variables of ECOG performance status (0 versus 1) and disease stage (recurrent versus primary metastatic). A permuted block of flexible size (2, 4) was used to generate the randomization allocation sequence. A new block was assigned to a stratum either when the first patient in that stratum was being randomized or when the randomization allocation sequence in the block assigned to that stratum had been used. Within one block, the randomization allocation sequence was assigned to patient numbers in order from the lowest to the highest. The random allocation sequences for patients and investigational drugs were generated and maintained by an independent, unblinded statistician from a third-party vendor, and were restricted to the study-blinded personnel, including the investigator, study center staff, patient, sponsor, clinical contract research organization and independent review committee. The random allocation sequence for investigational

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drugs was also held by the clinical supplies department of the sponsor, for the purpose of drug packaging.

TUMOR AND SAFETY ASSESSMENTS

Baseline tumor assessments included CT scans or MRIs of the nasopharynx, neck, chest and abdomen. A bone scan was to be performed if clinically indicated. All known sites of disease were documented at screening and re-assessed at each subsequent tumor evaluation. The same radiographic procedure was used throughout the study for each individual patient. Tumor assessments were performed every 6 weeks in the first 12 months and then every 9 weeks thereafter until disease progression, loss of clinical benefit, withdrawal of consent, initiation of new anti-cancer therapy or death, whichever occurred first. Clinical response was assessed by the investigator and BICR per RECIST v1.1.

Safety and tolerability were assessed in all patients who received any amount of study treatment. All adverse events were defined and graded according to the National Cancer Institute-Common Terminology Criteria for Adverse Events (NCI-CTCAE) version 5.0. After initiation of study drug administration, all adverse events, regardless of relationship to the study drug, were reported until 60 days after the last dose of the study drug. Any serious adverse events that occurred more than 60 days after the last dose of the study drug but were considered related to the study drug by investigator were also recorded. Immune-related adverse events (irAE) were reported as assessed by investigator.

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STATISTICAL METHODS

The methods outlined for PFS were also used for the OS and DoR analysis. The comparison of ORR and DCR between arms were analyzed using the stratified Cochran-Mantel-Haenszel test. The difference in ORRs between the two treatment arms and its associated 95% CI were determined using the stratified Mantel-Haenszel method. Safety analyses were performed on the safety population, which included all randomized patients who received any amount of study drug.

INTERIM ANALYSIS

By the data cutoff date of May 30, 2020, 128 BICR-assessed PFS events triggered the interim PFS analysis. On September 14, 2020, the iDMC reported that the efficacy boundaries for BICR-PFS and BICR-ORR had been crossed. The trial was unblinded following the iDMC's recommendation. Patients in the placebo group discontinued placebo treatment, whereas patients on active treatment in the toripalimab group continued to receive toripalimab.

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